4 BJT AC Analysis

INTRODUCTION

- In the previous chapter, we learnt how bipolar junction transistor works by giving DC supply. We also learned why BJT requires DC power supply.
- But the main question is, is there any use of BJT if it is operated only with DC supply?
  The answer is no.
  So, what is the use of BJT?
- The answer to this question gives the motivation to learners to learn BJT AC analysis which is more complicated than BJT DC analysis.
- We have discussed BJT as an amplifier in the previous chapter. The signal which is to be amplified is generally AC and small signal.
  The behaviour of BJT with AC signal applied at its input is very important.

Grey Matter Alert!

To use BJT as an amplifier, a DC power supply is must.

How BJT is used as an amplifier?

- In this section, we will apply AC as well as DC signal to a BJT and understand its amplification process.
  Consider a common emitter configuration of BJT.

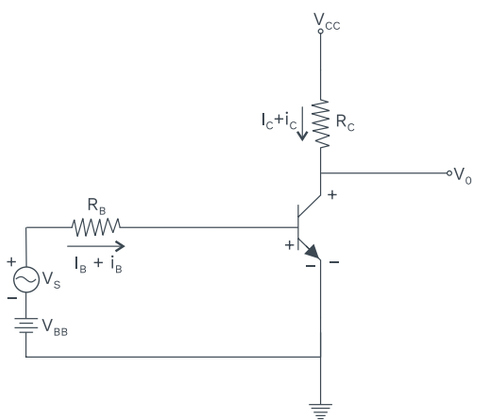


Fig. 4.1 CE Configuration with AC + DC Supply

- In the above circuit, both AC as well as DC inputs are applied. The AC signal is the main signal which is to be amplified.

Rack Your Brain

What is the meaning of the word transistor?

- If we consider V_S as sinusoidal signal voltage, then this time-varying voltage also produces time-varying base current i_B which is superimposed on quiescent base current.
- The time-varying base current also produces a time-varying collector current which is superimposed on quiescent collector current.
- This AC collector current induces a time-varying voltage across R_C, which finally induces an AC collector-emitter voltage.
- This AC collector-emitter voltage is the amplified version of small-signal voltage V_S.
- If i_B is time-varying then a time-varying voltage is also produced across base-emitter.

All the characteristics of BJT discussed are depicted in the graph below.

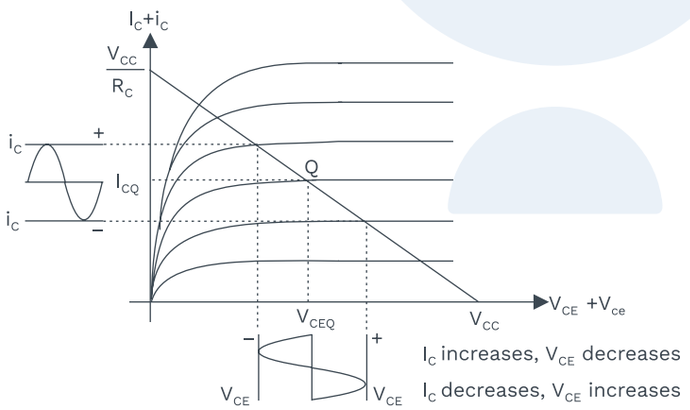


Fig. 4.2 CE Configuration Output Characteristics

Rack Your Brain

What is the DC operating point?

- The x-axis and y-axis not only contains DC voltage and current but also contains time-varying voltage and current.
- The time-varying base current induces time-varying voltage V_{be} and the relation between them is given by

$$I_B + i_B = \frac{I_S}{\beta} e^{\frac{V_{BE}+V_{be}}{V_T}}$$

Rack Your Brain

Why is the early effect important in BJT?

where I_S = reverse saturation current
V_T = thermal voltage

$$I_B + i_B = \frac{I_S}{\beta} \cdot e^{\frac{V_{BE}}{V_T}} \cdot e^{\frac{V_{be}}{V_T}}$$

$$I_B + i_B = I_B \, e^{\frac{V_{be}}{V_T}}$$

Using the Taylor series expansion, we get:

$$e^{\frac{V_{be}}{V_T}} = 1 + \frac{V_{be}}{V_T} \quad (V_T >> V_{be})$$

$$I_B + i_B = I_B \left(1 + \frac{V_{be}}{V_T}\right)$$

$$i_B = \frac{I_B \, V_{be}}{V_T}$$

The graph between i_B and V_{be} is shown below.

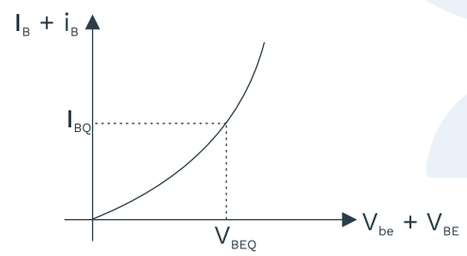


Fig. 4.3 Variation of Base Current

Rack Your Brain

What is small-signal analysis?

Grey Matter Alert!

BJT is not a linear device but can be considered as linear so that the superposition theorem can be applied.

SMALL-SIGNAL MODEL OF BJT

Various models exist which can be used to model a BJT to obtain various important parameters such as voltage gain, current gain, input impedance, output impedance with or without load.
There are three popular models for small-signal analysis of BJT:

- Hybrid equivalent model
- π–model
- r_e–model

Grey Matter Alert!

Hybrid parameters of a transistor gives real number at audio frequency range.

Each model can be obtained from other models. Let us start the discussion with hybrid equivalent model and later we will compare each model with a given BJT configuration.

Hybrid equivalent model

- BJT is a two port device with one port as a input, through which input is given to BJT and the other is output port through which amplified signal is taken as shown below.

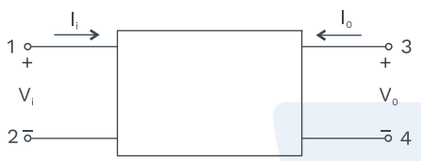


Fig. 4.4 Two-Port System

In hybrid parameter model, input voltage V_i at ports 1 and 2 and output current at ports 3 and 4 are represented in form of input current at ports 1 and 2 and output voltage at ports 3 and 4 respectively as:

$V_i = h_{11} I_i + h_{12} V_0$

$I_0 = h_{21} I_i + h_{22} V_0$

h_{11}, h_{12}, h_{21} and h_{22} are important parameters and will be discussed.

- **h_{11} (short circuit input impedance parameter)**
  h_{11} is the ratio of input voltage (V_i) and input current (I_i) with output terminal short circuit. Hence, it is also known as short circuit impedance parameter.

  It is measured in Ohm (Ω).

  It is given by

$$h_{11} = \left.\frac{V_i}{I_i}\right|_{V_0=0}$$

- **h_{12} (open circuit reverse transfer voltage ratio parameter)**
  It is the ratio of input voltage and output voltage with input terminal open circuit. It is

Rack Your Brain

What is the hybrid parameter?

Grey Matter Alert!

Open circuit impedance parameters and short circuit admittance parameters are very difficult to measure in transistors.

known as open circuit reverse transfer voltage ratio parameter.
It is a unitless parameter.
It is given by

$$h_{12} = \left.\frac{V_i}{V_0}\right|_{I_i=0}$$

- **h_{21} (short circuit forward current ratio parameter)**
  It is the ratio of output current to input current with output port short-circuited. It is also known as short circuit forward current ratio parameter.
  It is a unitless parameter.
  It is given by

$$h_{21} = \left.\frac{I_0}{I_i}\right|_{V_0=0}$$

- **h_{22} (open circuit output admittance parameter)**
  It is defined as the ratio of output current to output voltage with input terminal open-circuited. Hence it is known as the open circuit output admittance parameter.
  It is measured in Siemens and given by:

$$h_{22} = \left.\frac{I_0}{V_0}\right|_{I_i=0}$$

Complete hybrid equivalent model

After defining all the four parameters of the hybrid equivalent model such as:

$h_{11} \rightarrow$ input resistance $\rightarrow h_i$

$h_{12} \rightarrow$ reverse transfer voltage ratio $\rightarrow h_r$

$h_{21} \rightarrow$ forward transfer current ratio $\rightarrow h_f$

$h_{22} \rightarrow$ output conductance $\rightarrow h_o$

Input equation:

$$V_i = h_{11} I_i + h_{12} V_0$$

Grey Matter Alert!

Two back to back diodes can't function as BJT due to large base width.

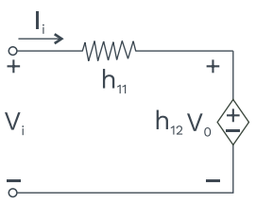


Fig. 4.5 Hybrid Input Equivalent Circuit

Output equation:

$I_0 = h_{21} I_i + h_{22} V_0$

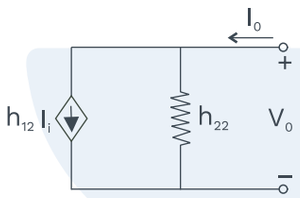


Fig. 4.6 Hybrid Output Equivalent Circuit

Combining the above two circuit, we get a complete hybrid equivalent model as shown below.

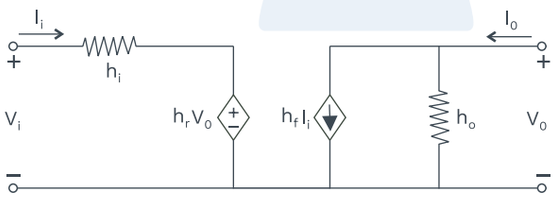


Fig. 4.7 Complete Hybrid Equivalent Circuit

BJT CONFIGURATIONS WITH HYBRID EQUIVALENT CIRCUIT

- In this section, we will discuss how BJT is replaced by its hybrid equivalent circuit.
- The BJT configuration model has the same format as the complete hybrid equivalent model with two terminals as input and output respectively and one terminal is common for both input and output terminals.

Common emitter configuration model

In common emitter configuration, the emitter is a common terminal with the input port being the base-emitter port and the output port being the collector-emitter port as shown below.

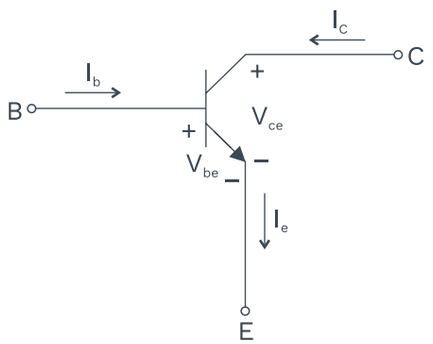


Fig. 4.8 Common Emitter Symbol

Rack Your Brain

Which BJT configuration is used as constant current source and why?

The hybrid equivalent circuit is shown below.

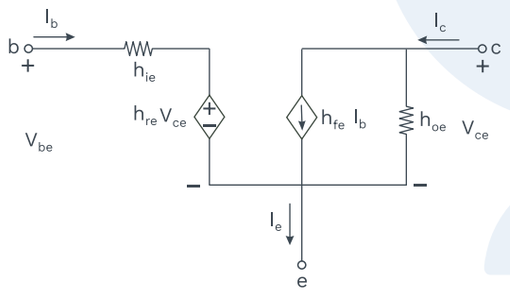


Fig. 4.9 CE Hybrid Equivalent Circuit

The subscript 'e' is used with 'h' to indicate common emitter.

Input current, $I_i = I_b$

$I_0 = I_c$

$I_e = I_b + I_c$

V_{be} = input voltage

V_{ce} = output voltage

Common base configuration model

In the common base configuration, base is the common terminal with the input port being the emitter-base and the output port being the collector base as shown below.

Grey Matter Alert!

The hybrid parameter model is used in analysis of BJT amplifier at medium or low frequency.

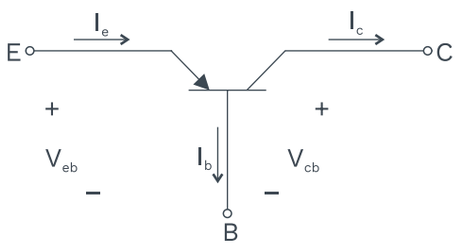


Fig. 4.10 Common Base Configuration

The equivalent circuit is shown below.

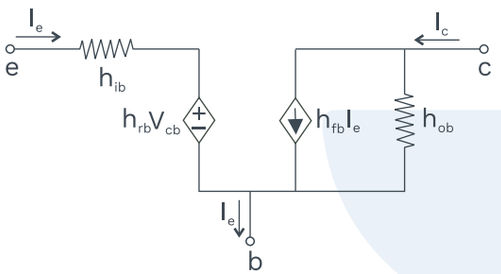


Fig. 4.11 Hybrid Equivalent Circuit

Example 1:
The signal which is applied to BJT for amplification is generally.

a) DC **b)** AC
c) AC + DC **d)** AC – DC

Sol:
The signal applied to BJT is AC + DC but for the amplification, generally AC signal is applied where DC is for the power supply.
Hence, the correct option is b).

Example 2:
h_{fe} in the hybrid equivalent model is:
a) Current gain
b) Voltage gain
c) Input impedance
d) Output impedance

Sol:
h_{fe} in the hybrid equivalent model is DC current gain.
Hence, the correct option is a).

Rack Your Brain

Why coupling capacitors are used in BJT amplifiers?

Example 3:
BJT transistor in a common emitter configuration has DC emitter current of 5 mA, current gain of 100, h_{oe} = 10 μS. Determine hybrid equivalent circuit.

Sol:
Small-signal AC emitter resistance is given by

$$r_e = \frac{26\ mV}{I_E}$$

$$r_e = \frac{26\ mV}{5\ mA}$$

$$r_e = 5.2\ \Omega$$

$$h_{ie} = \beta r_e = 100 \times 5.2$$

$$= 0.52\ k\Omega$$

$$r_0 = \frac{1}{h_{oe}} = \frac{1}{10\,\mu S}$$

$$r_0 = 100\ k\Omega$$

The common emitter equivalent model is given as:

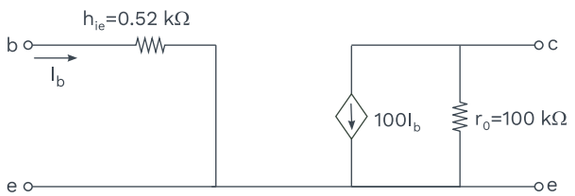


Fig. 4.12 Common Emitter Equivalent Model

BIASING CONFIGURATION OF THE HYBRID EQUIVALENT MODEL

Fixed-bias configuration

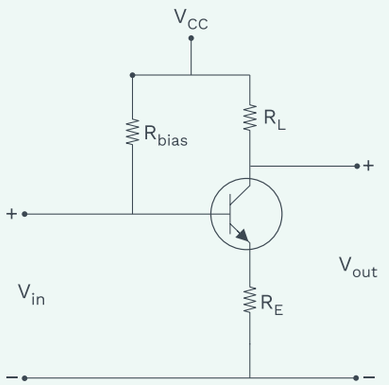
- DC analysis of fixed-bias configuration has been already discussed in the previous chapter, in this section we will discuss how AC signal is amplified using BJT and how it is applied and what are the other passive elements required for this circuit as shown below.

Grey Matter Alert!

Ohm meter cannot be used to measure the input impedance of an AC network.

Previous Years' Question

In the BJT amplifier shown in the figure is the transistor biased in the forward active region, putting a capacitor across R_E will:



a) Decrease the voltage gain and decrease the input impedance.
b) Increase the voltage gain and decrease the input impedance.
c) Decrease the voltage gain and increase the input impedance.
d) Increase the voltage gain and increase the input impedance.

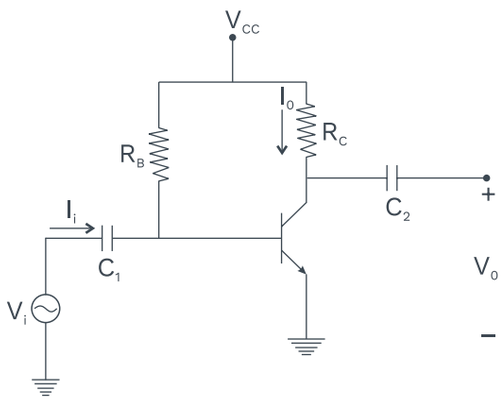


Fig. 4.13 Fixed-Bias Configuration

- C_1 and C_2 are coupling capacitors used mainly to couple low AC signal V_i to an amplifier.
- For DC signal C_1 and C_2 are open-circuited and short circuit for AC signal.

The small-signal model for the above circuit is shown below.

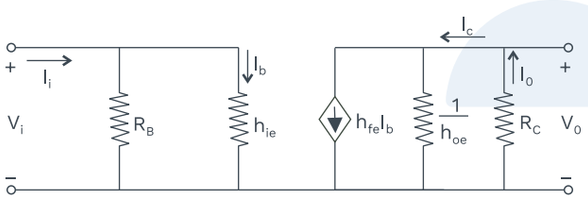


Fig. 4.14 Hybrid Equivalent Circuit of Fixed-Bias Configuration

Parameters of AC analysis

The important parameters for BJT AC analysis are:

- Input Impedance
- Output Impedance
- Voltage gain
- Current gain

Input impedance (Z_i)

It is the impedance seen from the input terminal V_i and is given by

$$Z_i = \frac{V_i}{I_i}$$

Rack Your Brain

What is the advantage of using a common emitter amplifier?

Grey Matter Alert!

Common base configuration provides highest voltage gain among all given configuration.

The current I_i is divided parallelly between R_B and h_{ie}:

$$V_i = I_i (R_B \parallel h_{ie})$$

$$Z_i = \frac{V_i}{I_i} = R_B \parallel h_{ie}$$

Output impedance (Z_0)

It is the impedance seen from the output terminal V_0 with input V_i being short circuit with $I_b = 0$, the output port is represented as shown below.

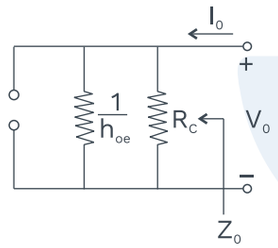


Fig. 4.15 Output Port of Fixed-Bias Circuit

The current I_0 is divided parallelly between $\frac{1}{h_{oe}}$ and R_C:

$$V_0 = I_0 \left(R_C \parallel \frac{1}{h_{oe}} \right)$$

$$\frac{V_0}{I_0} = R_C \parallel \frac{1}{h_{oe}}$$

$$Z_0 = R_C \parallel \frac{1}{h_{oe}}$$

Voltage gain (A_V)

It is the ratio of output voltage V_0 to input voltage V_i:

$$V_0 = -I_0 \left(\frac{1}{h_{oe}} \parallel R_C \right) = -I_C \left(\frac{1}{h_{oe}} \parallel R_C \right)$$

$$I_C = h_{fe} I_b$$

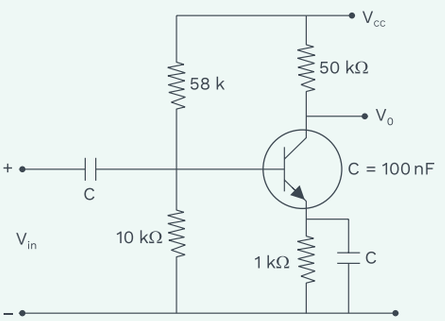
$$I_b = \frac{V_i}{h_{ie}}$$

Grey Matter Alert!

Common collector configuration provides the highest current gain among all given configurations.

Previous Years' Question

The transistor in the amplifier shown below has following parameters:
h_{fe} = 100, h_{ie} = 200 kΩ, h_{re} = 0, h_{oe} = 0.05 mho, C is very large. The output impedance is



a) 20 KΩ **b)** 16 KΩ
c) 5 KΩ **d)** 4 KΩ

$$V_0 = -h_{fe}\frac{V_i}{h_{ie}}\left(R_C \parallel \frac{1}{h_{oe}}\right)$$

$$\frac{V_0}{V_i} = -\frac{h_{fe}}{h_{ie}}\left(R_C \parallel \frac{1}{h_{oe}}\right)$$

$$A_V = \frac{V_0}{V_i} = -\frac{h_{fe}}{h_{ie}}\left(R_C \parallel \frac{1}{h_{oe}}\right)$$

Current gain (A_i)

It is the ratio of output current and input current and is given by

$$A_i = \frac{I_0}{I_i} = h_{fe}$$

Voltage divider configuration

After discussing fixed-bias configuration now we will discuss voltage divider configuration and its AC analysis with the small-signal model.

The voltage divider circuit with AC input is shown below.

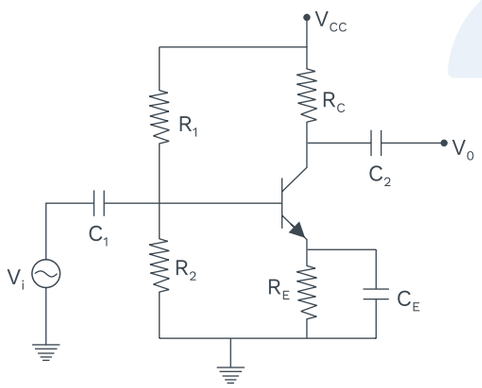


Fig. 4.16 Voltage Divider Configuration

DC analysis of the above has been already discussed in the last chapter, C_1 and C_2 are coupling capacitors.

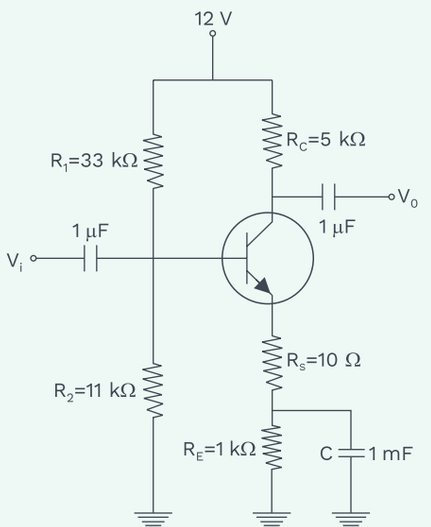
The capacitor C_E plays a very important role in the above circuit. With DC signal, is open-circuited increasing the input impedance and with AC signal C_E is short-circuited, the resistor R_E is bypassed, i.e. shorted, thus emitter is directly connected to the ground point.

Grey Matter Alert!

The common collector configuration has the lowest output impedance among all the given configurations.

Previous Years' Question

For the amplifier shown in the figure, the BJT parameters are V_{BE} = 0.7V, β=200, and thermal voltage V_T=25 mV. The voltage gain (V_0/V_i) of the amplifier is _____.



The above circuit is redrawn as shown below.

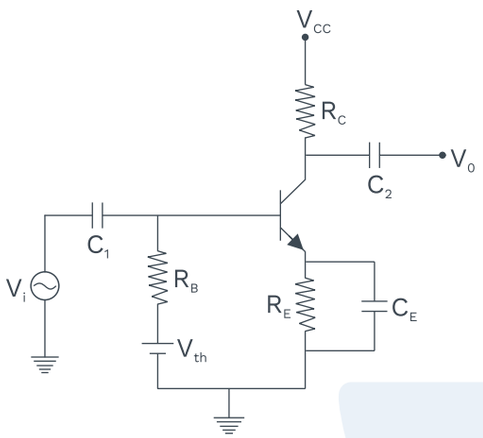


Fig. 4.17 Voltage Divider Bias Circuit

Base resistance, $R_B = \frac{R_1 R_2}{R_1 + R_2}$

Thevenin's voltage, $V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2}$

The small-signal equivalent circuit is shown below.

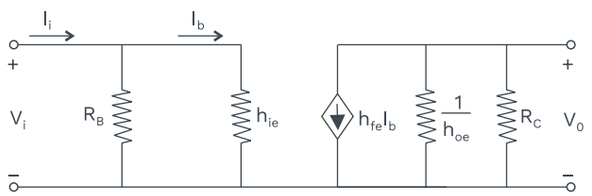


Fig. 4.18 Small-Signal Equivalent of Voltage Divider Bias

Input impedance (Z_i)

It is the ratio of V_i and I_i and is given as:

$$Z_i = \frac{V_i}{I_i} = R_1 \parallel R_2 \parallel h_{ie} = R_B \parallel h_{ie}$$

Output impedance (Z_0)

It is the ratio of V_0 and I_0 and is given as:

$$Z_0 = \frac{1}{h_{oe}} \parallel R_c$$

Rack Your Brain

What is the use of a bypass capacitor in BJT amplifier?

Grey Matter Alert!

Ohm meter cannot be used to measure the output impedance of an AC network.

If h_{oe} is very small,

$$Z_0 \cong R_C$$

Voltage gain (A_V)

Voltage gain of the above configuration is given by:

$$A_V = \frac{V_0}{V_i} = -\frac{h_{fe}}{h_{ie}}\left(R_C \parallel \frac{1}{h_{oe}}\right)$$

If $R_C \parallel \frac{1}{h_{oe}} \approx R_C$

$$A_V = \frac{V_0}{V_i} = -\frac{h_{fe}}{h_{ie}} R_C$$

Current gain (A_i)

Current gain is given by:

$$A_i = \frac{h_{fe}\,(R_1 \parallel R_2)}{(R_1 \parallel R_2) + h_{ie}}$$

or $$A_i = \frac{h_{fe}\,R_B}{R_B + h_{ie}}$$

Circuits with un-bypassed emitter-bias configuration

In the previous circuit, a bypassed capacitor is connected in parallel with resistance neglecting the effect of R_E when AC signal is applied. Now an un-bypassed resistor is used as shown below.

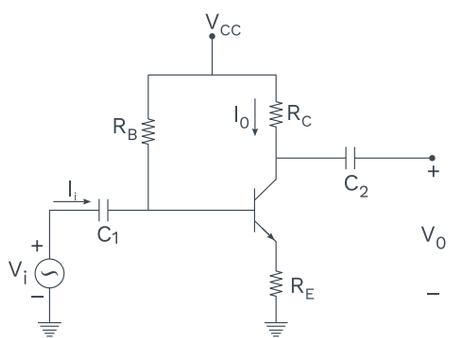


Fig. 4.19 Circuits with Un-Bypassed Emitter-Bias Configuration

Grey Matter Alert!

The common emitter is the most suitable configuration for voltage amplification.

Input impedance (Z_i)

Due to emitter resistance, the input impedance of configuration increases.

Base impedance, $Z_b = h_{fe}R_E$

Input impedance, $Z_i = R_B \parallel Z_b$

Output impedance (Z_0)

Since this is the common emitter configuration and the output is taken across collector, so there is no effect of R_E on output impedance.

$$Z_o = R_c$$

Voltage gain (A_V)

Voltage gain, $A_V = -\frac{h_{fe}R_C}{Z_b}$

$$A_V \cong -\frac{h_{fe}R_C}{h_{fe}R_E}$$

$$A_V \cong -\frac{R_C}{R_E}$$

Current gain (A_i)

$$A_i = -\frac{h_{fe}R_B}{R_B + Z_b}$$

Common collector configuration

After studying various biasing and models of common emitter, now we will study about common collector configuration model and its small-signal AC analysis.

The input impedance, output impedance, voltage gain, etc. will also be calculated for the common collector configuration.

Rack Your Brain

Why npn transistor is preferred over pnp?

Grey Matter Alert!

The common emitter configuration amplifiers is used in radio frequency transceiver circuits at IF stages.

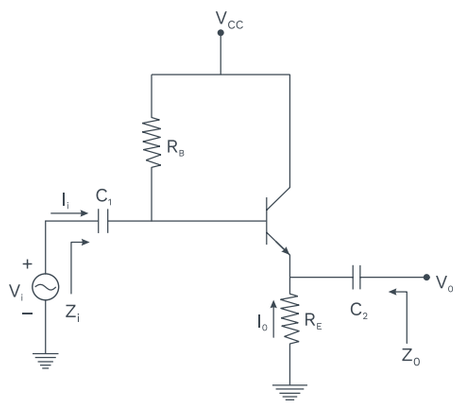


Fig. 4.20 Common Collector Configuration

In the above circuit, output V_0 is taken across emitter and collector is common to base and emitter.

Input impedance (Z_i)

Z_i is given by $Z_i = R_B \parallel Z_b$

Z_b = Impedance seen from the base terminal

$Z_b = h_{fe}R_E$

Output impedance (Z_0)

It is the impedance seen from the emitter terminal.

The output circuit, i.e. base-emitter circuit is redrawn as shown below.

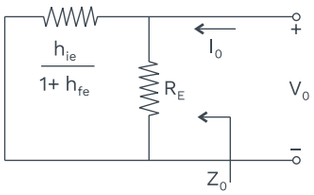


Fig. 4.21 Output Impedance Calculation

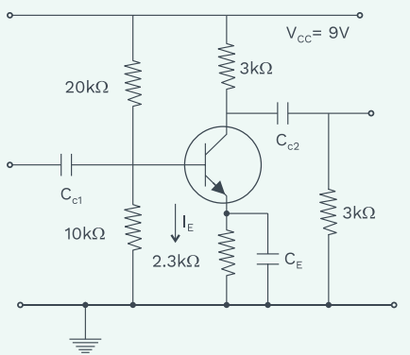
Common emitter current h_{fe} is very large, i.e. $h_{fe} + 1 \approx h_{fe}$

Output impedance, $Z_0 = \frac{V_0}{I_0} = R_E \parallel \frac{h_{ie}}{1+h_{fe}}$

or, $Z_0 = R_E \parallel \frac{h_{ie}}{h_{fe}}$

Previous Years' Question

In the following transistor circuit, V_{BE} = 0.7 V, r_e = 25 mV/I_E, β and all the capacitances are very large.



1) The Value of DC current I_E is:
a) 1 mA **b)** 2 mA
c) 5 mA **d)** 10 mA

2) The mid-band voltage gain of the amplifier is approximately:
a) –180 **b)** –120
c) –90 **d)** –60

Voltage gain (A_V)

It can be also calculated from the circuit given below.

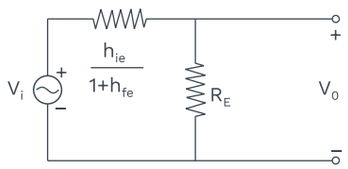


Fig. 4.22 Voltage Gain for CC Configuration

Voltage gain, $A_V = \frac{V_0}{V_i}$

By voltage division rule, we get:

$$V_0 = \frac{V_i \times R_E}{R_E + \left(\frac{h_{ie}}{1+h_{fe}}\right)}$$

$$A_V = \frac{V_0}{V_i} = \frac{R_E}{R_E + \left(\frac{h_{ie}}{h_{fe}}\right)} \qquad (1 + h_{fe} = h_{fe})$$

Current gain (A_i)

It is given by $A_i = \frac{h_{fe} R_B}{R_B + Z_b}$

Or $A_i = -A_V \frac{Z_i}{R_E}$

Common base configuration

After studying the parameters of common collector configuration, the common base h-parameter model is to be discussed.

In the common base configuration, the base is common to both input and output ports as shown below.

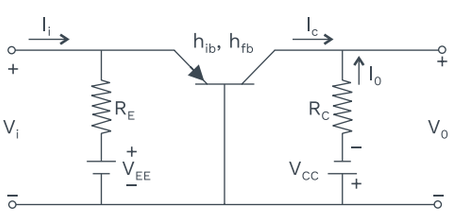


Fig. 4.23 Common Base Configuration

Rack Your Brain

What are the similarities between BJT and FET?

Grey Matter Alert!

Coupling capacitors are basically used to apply AC signal to BJT and isolate it from DC supply.

In the simplified h-parameter equivalent circuit of CB configuration, the parameter h_{rb} and h_{ob} are neglected. The small-signal model is shown below.

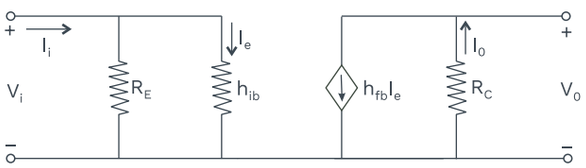


Fig. 4.24 Common Base Small-Signal Equivalent Parameters

Input impedance (Z_i)

It is the ratio of V_i and I_i.

Applying KVL in the input loop, we get:

$$V_i = I_i (R_E \parallel h_{ib})$$

$$Z_i = \frac{V_i}{I_i} = R_E \parallel h_{ib}$$

Output impedance (Z_0)

From the small-signal model, it is given as:

$$Z_0 = \frac{V_0}{I_0} = R_C$$

Voltage gain (A_V)

It is given by $A_V = \frac{V_0}{V_i}$

$$V_0 = - I_0 R_C = -(h_{fb})\, I_e R_C$$

$$I_e = \frac{V_i}{h_{ib}}$$

$$V_0 = -h_{fe} \frac{V_i}{h_{ib}} R_C$$

$$A_V = \frac{V_0}{V_i} = -\frac{h_{fe} R_C}{h_{ib}}$$

Current gain (A_i)

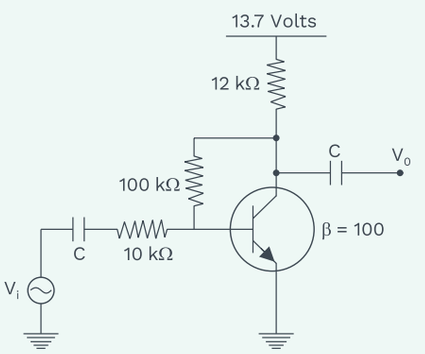
$$A_i = h_{fb} \cong -1$$

Grey Matter Alert!

The current gain is the least sensitive to changes in collector current.

Previous Years' Question

The voltage gain A_V, of the circuit shown below is



a) $|A_V| = 200$
b) $|A_V| = 100$
c) $|A_V| = 20$
d) $|A_V| = 10$

CONVERSION FORMULA FOR HYBRID PARAMETERS

The conversion formula provides easy method to obtain parameters of one configuration from the other given parameter.

Common base parameters in terms of common emitter is:

$$h_{ib} = \frac{h_{ie}}{1 + h_{fe}}$$

$$h_{fb} = -\frac{h_{fe}}{1 + h_{fe}}$$

$$h_{rb} = \frac{h_{ie}\, h_{oe}}{1 + h_{fe}} - h_{re}$$

$$h_{ob} = \frac{h_{oe}}{1 + h_{fe}}$$

Common collector parameters in terms of common emitter is:

$$h_{ic} = h_{ie}$$

$$h_{oc} = h_{oe}$$

$$h_{rc} = \frac{h_{ie}\, h_{oe}}{1 + h_{fe}} - h_{re}$$

$$h_{oc} = \frac{h_{oe}}{1 + h_{fe}}$$

COMPLETE HYBRID EQUIVALENT MODEL

The approximate hybrid equivalent model analysis gives us an idea about various parameters of transistors. In this section, we will derive equations for current gain, voltage gain, input impedance and output impedance in terms of h_i, h_r, h_f, h_o.

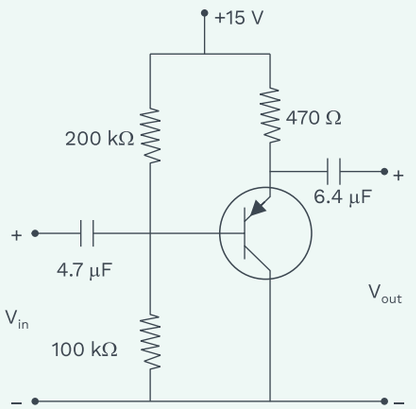
Consider the transistor as two-port system and it is shown below.

Grey Matter Alert!

The output impedance is the least sensitive parameter to the changes in temperature.

Previous Years' Question

For the amplifier circuit shown in the figure, the transistor has a β of 800. The mid band voltage gain V_0/V_i of the circuit will be



a) 0 **b)** < 1
c) = 1 **d)** 800

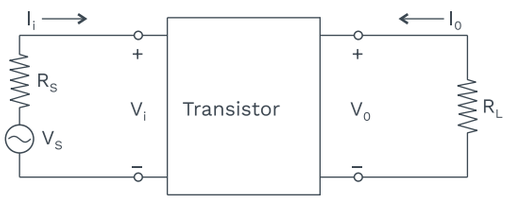


Fig. 4.25 Transistor as Two-Port Network

The complete hybrid equivalent circuit for the above two-port system is shown below.

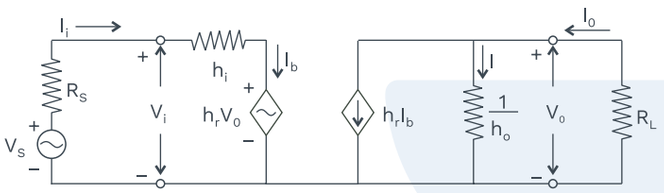


Fig. 4.26 Complete Hybrid Equivalent Model

Parameters calculation:

Current gain (A_i)

From the above circuit at the output node.

$$I_0 = h_f I_b + I$$

$$I_0 = h_f I_i + \frac{V_0}{\frac{1}{h_o}} \qquad (\because I_b = I_i)$$

$$I_0 = h_f I_i + h_o V_0$$

But $V_0 = -I_0 R_L$

$$I_0 = h_f I_i - h_o R_L I_0$$

$$I_0 + h_o R_L I_0 = h_f I_i$$

$$I_0 (1 + h_o R_L) = h_f I_i$$

$$A_i = \frac{I_0}{I_i} = \frac{h_f}{1 + h_o R_L}$$

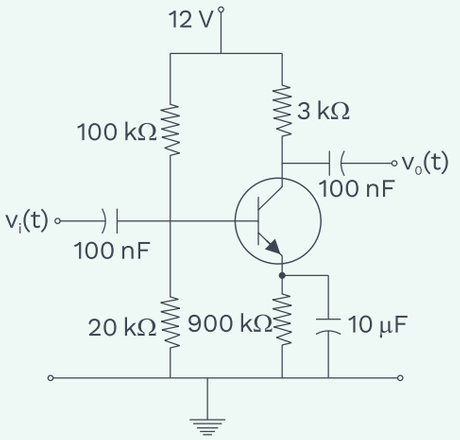
This current gain expression is different from that obtained from the approximate model.

Voltage gain (A_V)

It is the ratio of V_0 and V_i, i.e. $A_V = \frac{V_0}{V_i}$

Previous Years' Question

A small-signal source $v_i(t) = A \cos 20\ t + B \sin 10^6\ t$ is applied to a transistor amplifier as shown below. The transistor has $\beta = 150$ and $h_{ie} = 2\ k\Omega$. Which expression best approximates $v_0(t)$?



a) $v_0(t) = -1500(A\cos 20t + B\sin 10^6 t)$
b) $v_0(t) = -150(A\cos 20t + B\sin 10^6 t)$
c) $v_0(t) = -1500\ B \sin 10^6 t$
d) $v_0(t) = -150\ B \sin 10^6 t$

From the input loop, we get:

$$V_i = I_i h_i + h_r V_0$$

From current gain, $I_i = \frac{I_0\left(1+h_o R_L\right)}{h_f}$

also $I_0 = -\frac{V_0}{R_L}$

$$V_i = -\frac{\left(1+h_o R_L\right)}{h_f R_L} h_i V_0 + h_r V_0$$

$$\frac{V_0}{V_i} = -\frac{h_f R_L}{h_i + \left(h_i h_o - h_f h_r\right) R_L}$$

Voltage gain $A_V = -\frac{h_f R_L}{h_i + \left(h_i h_o - h_f h_r\right) R_L}$

Input impedance (Z_i)

It is the ratio of input voltage to input current.

$$Z_i = \frac{V_i}{I_i}$$

$$V_i = h_i I_i + h_r V_0$$

$$V_0 = -I_0 R_L$$

$$V_i = h_i I_i - h_r R_L I_0$$

$$A_i = \frac{I_0}{I_i}$$

$$I_0 = A_i I_i$$

$$V_i = h_i I_i - h_r R_L A_i I_i$$

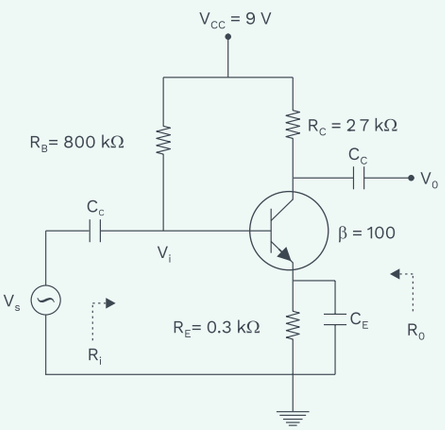
$$Z_i = \frac{V_i}{I_i} = h_i - h_r R_L A_i$$

Current gain, $A_i = \frac{h_f}{1+h_o R_L}$

$$Z_i = \frac{V_i}{I_i} = h_i - \frac{h_f h_r R_L}{1+h_o R_L}$$

Previous Years' Question

The amplifier circuit shown below uses a silicon transistor. The capacitors C_C and C_E can be assumed to be short at signal frequency and the effect of output resistance r_0 can be ignored. If C_E is disconnected from the circuit, which one of the following statements is TRUE?



a) The input resistance R_i increases and the magnitude of voltage gain A_V decreases.

b) The input resistance R_i decreases and the magnitude of voltage gain A_V increases.

c) Both input resistance R_i and the magnitude of voltage gain A_V decrease.

d) Both input resistance R_i and the magnitude of voltage gain A_V increases.

Input impedance, $Z_i = h_i - \frac{h_f h_r R_L}{1 + h_o R_L}$

Output impedance (Z_0)

It is defined as the ratio of output voltage to output current.

$$Z_0 = \frac{V_0}{I_0}$$

For output impedance, V_s should be shorted.

$$I_i = \frac{-h_r V_0}{R_s + h_i}$$

$$I_0 = h_f I_i + h_o V_0$$

$$= \frac{-h_f h_r V_0}{R_s + h_i} + h_o V_0$$

$$\frac{V_0}{I_0} = \frac{1}{\left[h_o - \left(\frac{h_f h_r}{(h_i + R_s)}\right)\right]}$$

Output impedance is:

$$Z_0 = \frac{1}{\left[h_o - \left(\frac{h_f h_r}{h_i + R_s}\right)\right]}$$

Typical h-parameters values for transistor

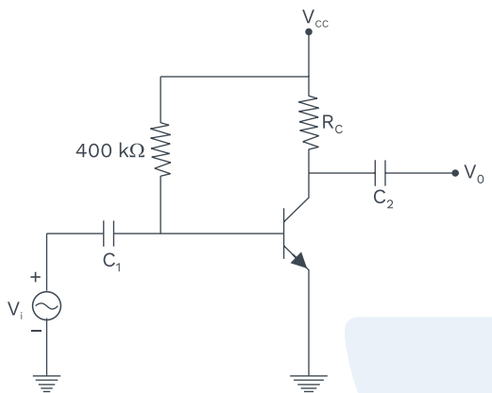
There are some approximate values of h-parameter which is very useful in calculations.

Parameters	CE	CC	CB
$h_{11} = h_i$	1100 Ω	1100 Ω	21.6 Ω
$h_{12} = h_r$	2.5×10^{-4}	≅ 1	2.9×10^{-4}
$h_{21} = h_f$	50	−51	−0.98
$h_{22} = h_o$	24 μA/V	25 μA/V	0.49 μA/V
$1/h_o$	40 kΩ	40 kΩ	2 MΩ

Let us discuss some examples to understand how to apply h-model circuit to calculate the parameters of transistor.

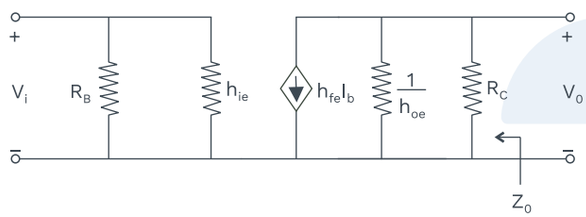
Example 4:

A fixed-bias circuit is shown below, calculate input impedance using the h-parameter model.



Sol:

Approximate model of the above circuit can be redrawn as shown below.



Typical value of h_{ie} = 1100 Ω

h_{ie} = 1.1 kΩ

R_B = 400 kΩ

Input impedance $Z_i = R_B || h_{ie}$.

Z_i = (400||1.1) kΩ

Z_i = 1.096 kΩ

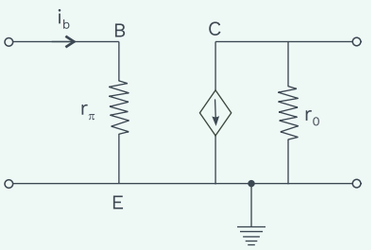
Note:

i) We can see that 1.096 kΩ is very close to 1.1 kΩ.

ii) For AC analysis model, C_1 and C_2 are short-circuited.

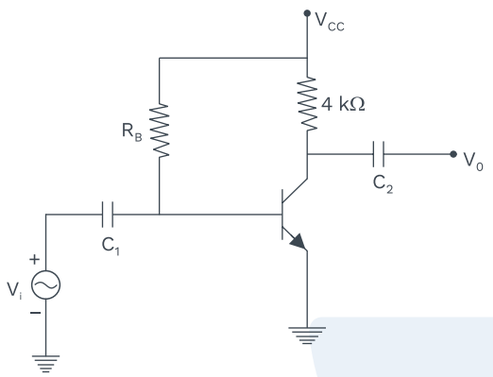
Previous Years' Question

The current i_D through the base of a silicon npn transistor is 1 + 0.1 cos (10000πt)mA. At 300 K, the r_π in the small-signal model of the transistor is:



a) 250 Ω **b)** 27.5 Ω
c) 25 Ω **d)** 22.5 Ω

Example 5:
A fixed-bias configuration circuit is given below, calculate output impedance from hybrid model.

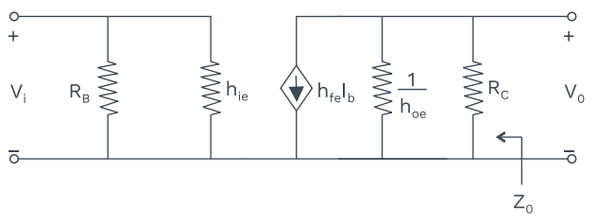


Sol:
Output impedance of the above circuit is given by

$$Z_0 = \frac{1}{h_{oe}} \| R_C$$

$R_C = 4k\Omega$

Equivalent model is given by



$h_{oe} = 24\ \mu A/V$

$$\frac{1}{h_{oe}} = \frac{1}{24 \times 10^{-6}} = 41.66\ k\Omega$$

$Z_0 = (41.66||4)\ k\Omega$

$Z_0 = 3.65\ k\Omega$

Previous Years' Question

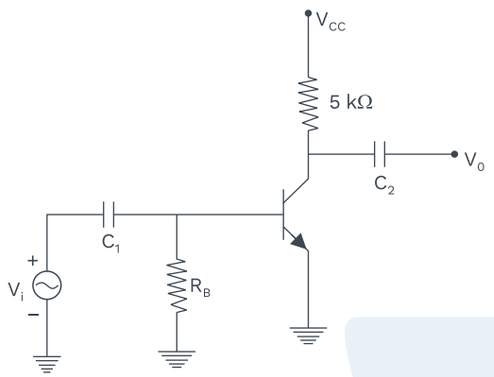
The quiescent collector current I_C of a transistor is increased by changing resistances. As a result:

a) g_m will not be affected
b) g_m will decrease
c) g_m will increase
d) g_m will increase or decrease depending upon bias stability.

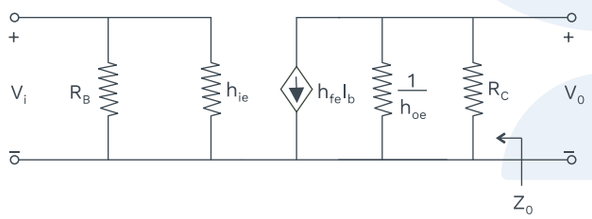
Rack Your Brain

What is the AC load line?

Example 6:
For the circuit shown below, what is the voltage gain for typical values of h-parameters.



Sol:
Hybrid equivalent model is given by



Voltage gain for the above circuit is given by

$$A_V = -h_{fe}\frac{\left(R_C \parallel \left(\frac{1}{h_{oe}}\right)\right)}{h_{ie}}$$

$h_{fe} = 50$

$h_{oe} = 24\ \mu A/V$

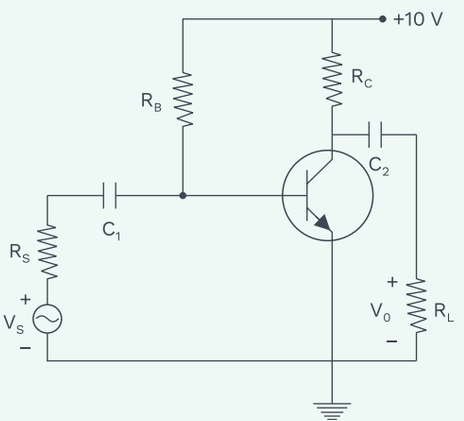
$h_{ie} = 1.1\ k\Omega$

$$A_V = -50\frac{(5 \parallel 41.66)}{1.1}$$

$A_V = -202.9 \approx -203$

Previous Years' Question

Consider the common emitter amplifier shown below with the following circuit parameters:
$\beta = 100$, $g_m = 0.3861$ A/V, $r_0 = \infty$, $r_\pi = 259\ \Omega$, $R_S = 1\ k\Omega$, $R_B = 93\ \Omega$, $R_C = 250\ \Omega$, $R_L = 1\ k\Omega$, $C_1 = \infty$ and $C_2 = 4.7\ \mu F$.



1) The resistance seen by source V_S is:
a) 250 Ω **b)** 1258 Ω
c) 93 kΩ **d)** ∞

2) The lower cut-off frequency due to C_2 is:
a) 33.9 HZ **b)** 27.1 Hz
c) 13.6 Hz **d)** 16.9 Hz

Example 7:
The voltage gain of the common emitter configuration provides phase shift of:

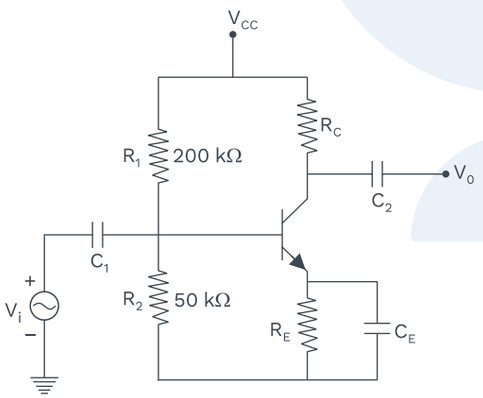
a) 0° **b)** 180°
c) 360° **d)** 90°

Sol:
The expression for the voltage gain of the common emitter configuration contains negative sign which indicates that there is a phase shift of 180° between input and output.
Hence, the correct option is b).

Grey Matter Alert!

π-model is a linear two-port network model of BJT using small-signal base-emitter voltage and collector-emitter as independent variables.

Example 8:
A voltage divider bias configuration is given below, the input impedance for the circuit is:



Grey Matter Alert!

The common collector is also known as emitter follower configuration.

Sol:
Given configuration is a voltage divider configuration.
DC analysis:
Base resistance = Thevenin's resistance

$$R_B = R_{Th}$$

$$R_B = (200 \parallel 50)\ k\Omega$$

$$R_B = \left(\frac{200k \times 50k}{250k}\right)$$

$$R_B = 40\ k\Omega$$

$$V_B = V_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{V_{CC}50k}{250k}$$

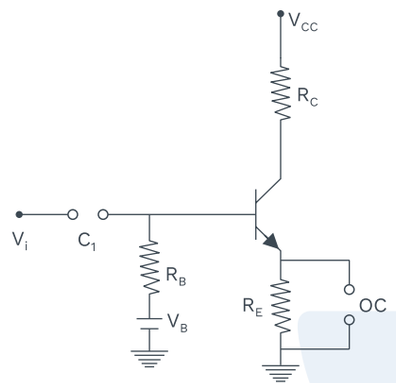
Previous Years' Question

The current gain of a BJT is:

a) $g_m r_0$ **b)** g_m/r_0
c) $g_m r_\pi$ **d)** g_m/r_π

$$V_B = \frac{V_{CC}}{5}$$

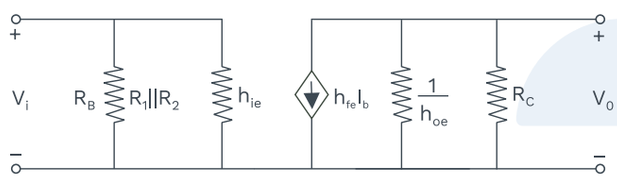
The circuit can be redrawn as shown below.



The capacitor is replaced by an open circuit.

AC analysis:

The hybrid equivalent model is given by



h_{ie} = 1.1 kΩ

$R_B = R_1 \| R_2$ = 40 kΩ

Input impedance $Z_i = R_1 \| R_2 \| h_{ie}$

Z_i = (40 || 1.1) kΩ

$$Z_i = \left(\frac{40 \times 1.1}{41.1}\right) k\Omega$$

Z_i = 1.07 kΩ

Rack Your Brain

What happens if we do not give DC power supply to the transistor?

Rack Your Brain

What is the high-frequency π-model?

HYBRID π-MODEL

- The next model to be discussed is the π-model. There are various advantages of π-model over the hybrid model which will be very helpful in the analysis of transistor amplifier.
  To understand π-model, consider a two port BJT as shown below.

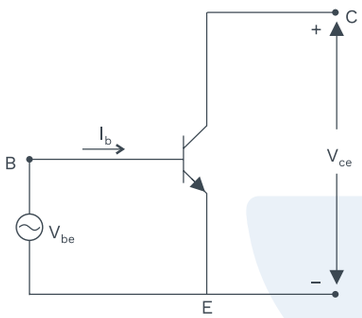


Fig. 4.27 BJT Amplifier with Only Small-Signal

- In the above circuit, the signal to be amplified is applied between base and emitter is V_{be}.
- Due to V_{be}, a base current i_b also flows through the base.
- Now, since i_b is flowing due to V_{be}, so there must be a relation between V_{be} and i_b according to Ohm's law.
  As V_{be} is high, i_b is high which implies that

$$V_{be} \propto i_b$$

$$V_{be} = r_\pi \, i_b$$

$$r_\pi = \frac{V_{be}}{i_b}$$

  r_π is known as the small-signal resistance.
- Emitter-base junction is similar to a two terminal diode element with the base as anode and emitter as cathode with characteristics shown below.

Rack Your Brain

When a two-port network is called as symmetrical or reciprocal?

Grey Matter Alert!

Common collector fixed-bias configuration has significant voltage gain and it's input impedance is low.

Rack Your Brain

How does the common emitter amplifier provides 180° phase shift between the input and output?

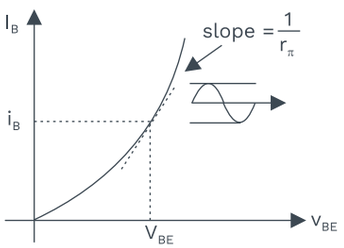


Fig. 4.28

r_π is the inverse of slope of above characteristics and is given by:

$$\frac{1}{r_\pi} = \frac{\partial i_B}{\partial V_{BE}}$$

$$= \frac{\partial}{\partial V_{BE}}\left[\frac{I_S}{\beta} e^{\frac{V_{BE}}{V_T}}\right]_{Q\text{–point}}$$

$$\frac{1}{r_\pi} = \frac{1}{V_T}\left[\frac{I_S}{\beta} e^{\frac{V_{BE}}{V_T}}\right]$$

$$\frac{1}{r_\pi} = \frac{I_B}{V_T}$$

$$\frac{V_{be}}{i_b} = r_\pi = \frac{V_T}{I_B}$$

$$I_B = \frac{I_C}{\beta}$$

$$r_\pi = \frac{\beta V_T}{I_C}$$

- Due to the small-signal V_{be}, there is small change in the collector current i_c which is given by

$$\Delta i_c = \frac{\partial i_c}{\partial V_{BE}} \cdot \Delta V_{BE}$$

$$i_c = \frac{\partial i_c}{\partial V_{BE}}$$

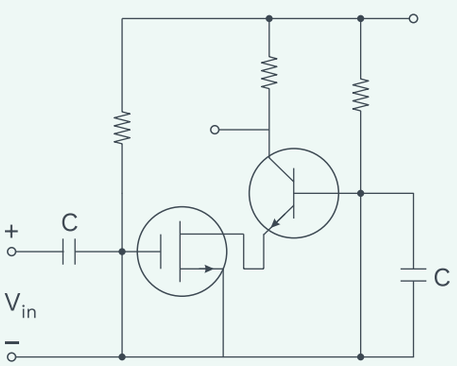
or $i_c = I_s e^{(V_{BE}/V_T)}$

Grey Matter Alert!

The voltage divider configuration has higher stability factor than fixed-bias configuration.

Previous Years' Question

The amplifier circuit shown below, uses a composite transistor of a MOSFET and BIPOLAR in cascade. All capacitance are large. g_m of the MOSFET = 2 mA/V, and h_{fe} of the BIPOLAR = 99. The overall Transconductance g_m of the composite transistor is



a) 198 mA/V
b) 9.9 mA/V
c) 4.95 mA/V
d) 1.98 mA/V

$$\frac{\partial i_c}{\partial V_{BE}} = \frac{1}{V_T} \cdot I_S \, e^{(V_{BE}/V_T)} \Big|_{Q\text{-point}}$$

$$\frac{\partial i_c}{\partial V_{BE}} = \frac{I_C}{V_T}$$

I_C is nothing but the Q-point and ratio I_C/V_T is known a transconductance.

Transconductance, $g_m = \frac{I_C}{V_T}$

With the help of r_π and g_m, we will develop a new model known as the hybrid π-model as shown below.

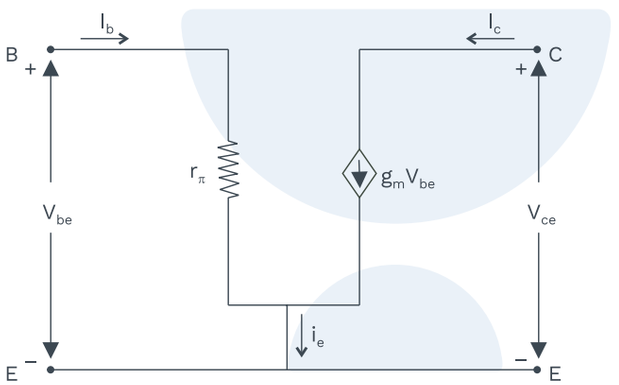


Fig. 4.29 Hybrid π-Equivalent of BJT

DIFFERENT BJT CONFIGURATIONS USING π-MODEL

- In hybrid equivalent circuit also, we discuss all the configurations of BJT using the hybrid equivalent. In this section, we discuss about different BJT configuration using π-model.
- For AC analysis, we will follow the same steps as we did in the case of the hybrid equivalent model for the analysis.
- Replace the transistor by its equivalent circuit.
- All DC voltages should be short-circuited.
- All coupling and bypass capacitors are short-circuited at AC.

Common emitter circuit

The transistor parameter obtained using hybrid model such as,

- Input impedance
- Output impedance

- Voltage gain
- Current gain

Consider a common emitter BJT with voltage divider bias configuration.

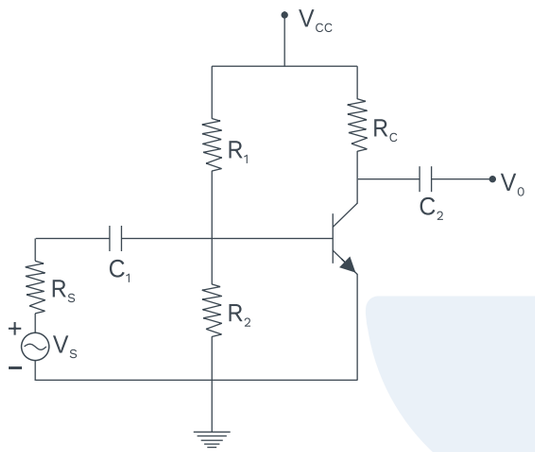


Fig. 4.30 CE Configuration

g_m and r_π calculation

g_m and r_π can be calculated by performing DC analysis.

Remove capacitors by open circuit and calculate Thevenin's voltage and resistance.

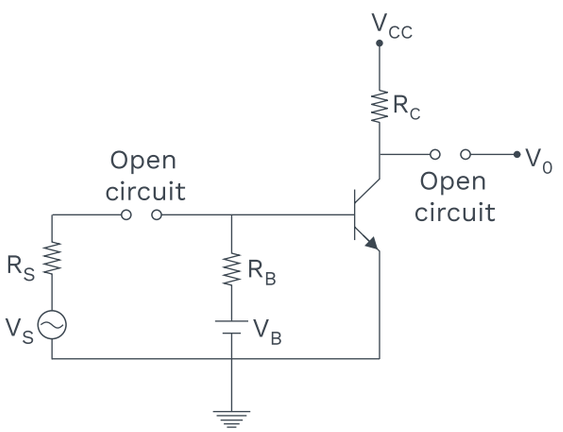


Fig. 4.31 DC Analysis of CE

$$V_B = \frac{V_{CC}R_2}{R_1 + R_2}, \quad R_B = \frac{R_1R_2}{R_1 + R_2}$$

Grey Matter Alert!

The common base is commonly used as a cascode stage to isolate output voltage signal from feeding back to input.

$$g_m = \frac{I_C}{V_T} \text{ transconductance}$$

$$r_\pi = \frac{V_T}{I_b} \text{ small-signal resistance}$$

After calculating g_m and r_π, the small-signal equivalent model is given as

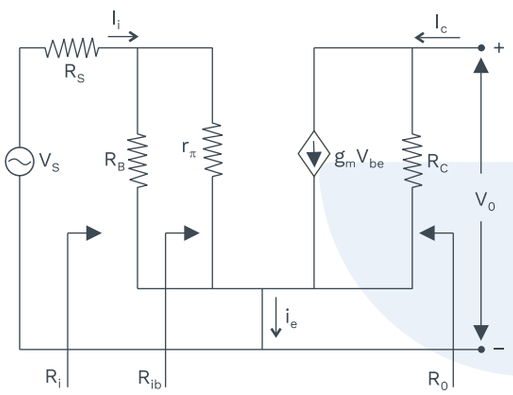


Fig. 4.32 Small-Signal π-Model of CE Configuration

- **Input impedance (Z_i)**
  It is given by V_s/I_i
  Since I_i is divided parallelly between R_B and r_π
  Small-signal base resistance is $R_{ib} = r_\pi$
  Input impedance $Z_i = R_B \parallel R_{ib}$

  $Z_i = R_1 \parallel R_2 \parallel R_{ib}$ or $Z_i = R_1 \parallel R_2 \parallel r_\pi$

- **Output impedance (Z_0)**
  For output impedance V_s should be shorted.
  For $V_s = 0$

  $I_b = 0 \Rightarrow I_C = 0 \Rightarrow g_m = 0$

  $$\frac{V_0}{I_0} = R_C$$

- **Voltage gain (A_V)**
  Output voltage $V_0 = -g_m V_{be} R_C$
  or $V_0 = -(\beta I_b) R_C$

  $V_{in} = I_b r_\pi$

Grey Matter Alert!

The Darlington pair amplifier uses two cascode amplifiers.

$$V_{in} = \frac{V_S R_i}{R_i + R_S}$$

$$A_V = \frac{V_0}{V_S} = -\frac{\beta R_c i_b}{V_S} = -\beta R_C \frac{V_{in}}{R_{ib}} \frac{1}{V_S}$$

$$A_V = -\frac{\beta R_C}{r_\pi} \frac{R_i}{R_i + R_S}$$

Common emitter configuration with emitter resistance

A common emitter circuit with voltage divider bias and emitter resistance R_E is shown below.

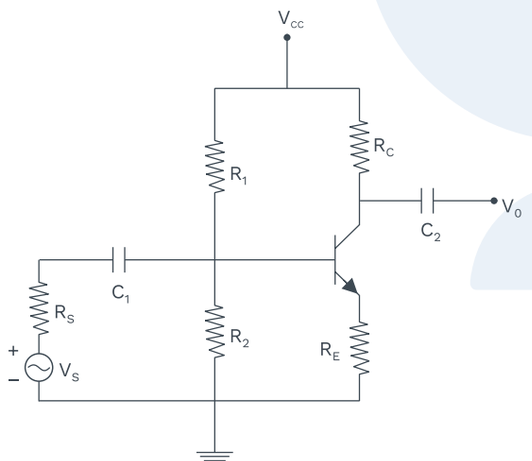


Fig. 4.33 CE Configuration with Resistance R_E

Resistance R_E is connected which increases the input resistance. The π-equivalent model for the above configuration is shown below.

Grey Matter Alert!

The voltage gain of common collector configuration is always less than unity.

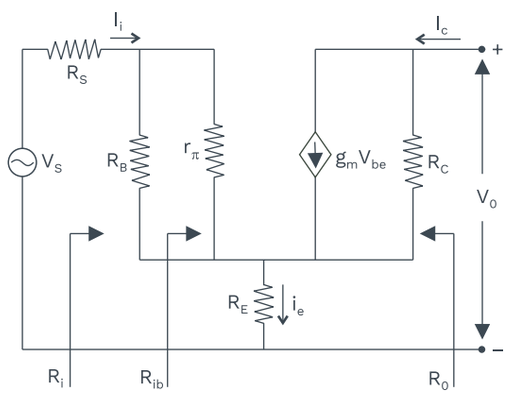


Fig. 4.34 Small-Signal π-Model of CE Configuration with Emitter Resistance R_E

Grey Matter Alert!

The common collector configuration is perfect for driving heavy loads.

- **Input impedance (Z_i)**
  It is given as $R_i = R_1 \| R_2 \| R_{ib}$

$$R_{ib} = \frac{V_{in}}{I_b} = r_\pi + (1+\beta)R_E$$

$$R_i = R_1 \| R_2 \| R_{ib}$$

- **Output impedance (Z_0)**
  Since the output of the amplifier circuit is taken across collector, so it remains same as R_C.

$$Z_0 = R_C$$

- **Voltage gain (A_V)**

$$V_0 = -\beta I_b R_C$$

$$V_{in} = I_b r_\pi + (I_b + \beta I_b)R_E$$

$$V_{in} = \frac{V_s \cdot R_i}{R_i + R_s}$$

$$\text{Voltage gain } \frac{V_0}{V_s} = -\frac{\beta I_b R_C}{V_s}$$

$$= -\beta R_C \frac{V_{in}}{R_{ib}} \frac{1}{V_S}$$

$$A_V = \frac{-\beta R_C}{r_\pi + (1+\beta)R_E}\left(\frac{R_i}{R_i + R_s}\right)$$

Common base amplifier π-model

The analysis of common base amplifier has been done using a hybrid model. Also, we had derived the expressions for input impedance, output impedance, voltage gain and current gain and the same will be obtained by using π-model. Consider a common base amplifier as shown below.

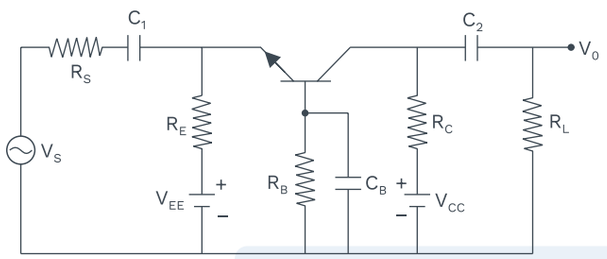


Fig. 4.35 Common Base Amplifier

In a common base amplifier, the output is taken across the collector terminal, C_B is the base capacitance.

The small-signal π-model of the above circuit is shown below.

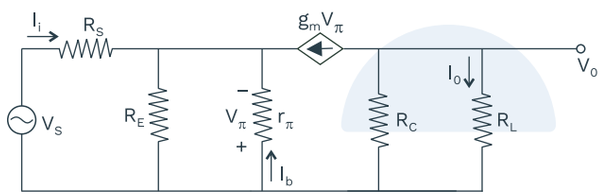


Fig. 4.36 π-Model for Common Base Amplifier

- **Voltage gain (A_V)**

  V_0 is given by $V_0 = -g_m V_\pi (R_C \| R_L)$

  Applying KCL at emitter node in small-signal equivalent circuit, we get:

$$g_m V_\pi + \frac{V_\pi}{r_\pi} + \frac{V_\pi}{R_E} + \frac{V_S - (-V_\pi)}{R_S} = 0$$

  But $\beta = g_m r_\pi$

$$V_\pi \left(g_m + \frac{1}{r_\pi} + \frac{1}{R_E} + \frac{1}{R_S} \right) = -\frac{V_S}{R_S}$$

$$V_\pi = -\frac{V_S}{R_S} \left[\left(\frac{r_\pi}{1+\beta} \right) \| R_E \| R_S \right]$$

Putting the value of V_π in the first equation, we get:

$$A_V = \frac{V_0}{V_S} = g_m \left(\frac{R_C \,||\, R_L}{R_S} \right) \left[\left(\frac{r_\pi}{1+\beta} \right) || \, R_E \,||\, R_S \right]$$

If R_S is low (in ohms) as compared to other resistances, then:

$$A_V = g_m \left(R_C \,||\, R_L \right)$$

- **Current gain (A_i)**

It is the ratio given between I_0 and I_i.

Applying KCL at the emitter node, we get:

$$I_i + \frac{V_\pi}{r_\pi} + g_m V_\pi + \frac{V_\pi}{R_E} = 0$$

$$V_\pi \left(\frac{1}{r_\pi} + g_m + \frac{1}{R_E} \right) = -I_i$$

$$V_\pi = -I_i \left[\left(\frac{r_\pi}{1+\beta} \right) || \, R_E \right]$$

$$I_0 = -(g_m V_\pi) \left(\frac{R_C}{R_C + R_L} \right)$$

Current gain,

$$A_i = \frac{I_0}{I_i} = g_m \left(\frac{R_C}{R_C + R_L} \right) \left[\frac{r_\pi}{(1+\beta)} || \, R_E \right]$$

- **Input impedance (Z_i)**

Since this is a common base configuration, the impedance between emitter terminal and base terminal is known as input impedance. The emitter-base circuit is redrawn as shown below.

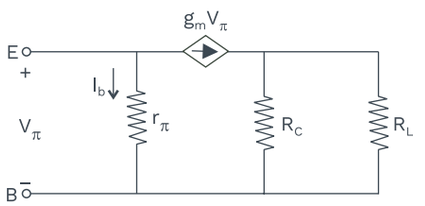


Fig. 4.37 Input Circuit of Common Base

Resistance looking at emitter terminal is:

$$R_{ie} = \frac{V_\pi}{I_i}$$

Applying KCL at emitter node, we get:

$$I_i = I_b + g_m V_\pi = \frac{v_\pi}{r_\pi} + g_m V_\pi = V_\pi \left(\frac{1+\beta}{r_\pi} \right)$$

$$R_{ie} = \frac{V_\pi}{I_i} = \frac{r_\pi}{1+\beta}$$

- **Output impedance (Z_0)**

Output impedance can be calculated by looking at the collector-base port as shown below.

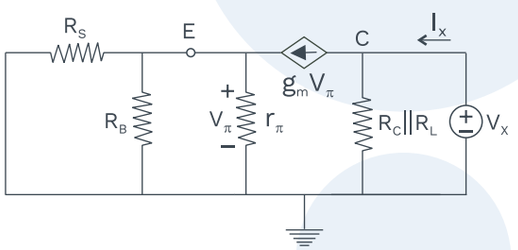


Fig. 4.38 Output Circuit of Common Base

Applying KCL at emitter node, we get:

$$g_m V_\pi + \frac{V_\pi}{r_\pi} + \frac{V_\pi}{R_E} + \frac{V_\pi}{R_S} = 0$$

V_s is shorted, so $V_\pi = 0$

So, the dependent current source is open circuit and current I_x flows only through R_c and $R_{L.}$

$\therefore$ Output impedance, $Z_0 = \frac{V_x}{I_x} = R_C \| R_L$

$$Z_0 = R_C \| R_L$$

Common-collector configuration amplifier π-model

We have discussed common emitter and common base amplifier using π-model which are very useful for high-frequency signals.

Consider a common-collector amplifier as shown below.

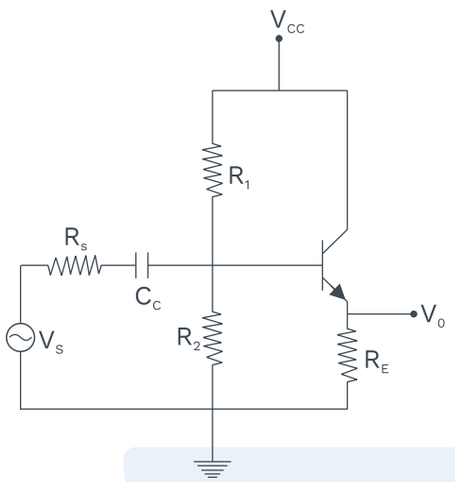


Fig. 4.39 Common-Collector Circuit.

V_s is the small-signal to be amplified and R_E is the emitter resistance across which output is taken.
Small-signal equivalent circuit is shown below.

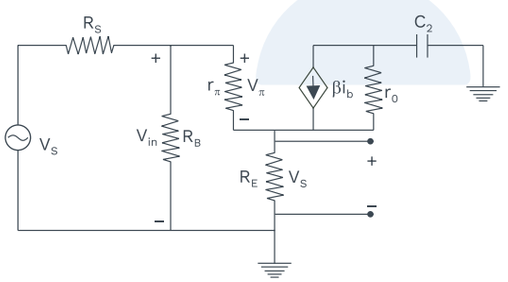


Fig. 4.40 Small-Signal Equivalent of Common-Collector Amplifier

The parameters for this configuration is same as that calculated for other configuration such as:

- Input impedance
- Output impedance
- Voltage gain
- Current gain

For the calculation of above parameters, the small-signal circuit can be redrawn as shown below.

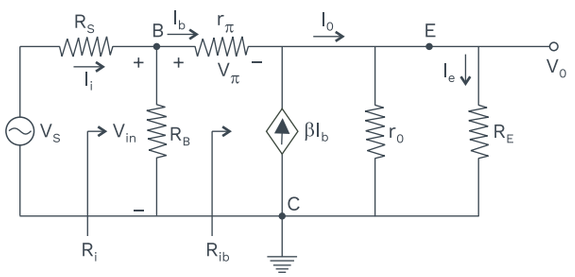


Fig. 4.41 Small-Signal Common Collector Amplifier Modified Circuit

- **Voltage gain (A_V)**
  It is the ratio of 'V_0' and 'V_s'
  I_b is current through r_π

  Output current,

  $$I_0 = i_b + \beta I_b$$

  $$I_0 = i_b(1 + \beta)$$

  Output voltage $V_0 = I_b(1 + \beta)(r_0 \| R_E)$
  Applying KVL in the base-emitter loop, we get:

  $$V_{in} = I_b[r_\pi + (1 + \beta)(r_0 \| R_E)]$$

  Using voltage division rule, we get:

  $$V_{in} = \frac{V_s R_i}{R_i + R_s}$$

  Where $R_i = R_1 \| R_2 \| R_{ib}$

  Small-signal voltage gain, $A_v = \frac{V_0}{V_s}$

  $$A_v = \frac{(1+\beta)(r_0 \| R_E)}{r_\pi + (1+\beta)(r_0 \| R_E)} \left(\frac{R_i}{R_i + R_s} \right)$$

- **Input impedance (Z_i)**
  It is the impedance looking into base terminal. It is R_{ib} given as:

  $$R_{ib} = r_\pi + (1+\beta)(r_0 \| R_E)$$

- **Output impedance (Z_0)**
  Replace V_s by short circuit and redraw the circuit with V_x and I_x at output terminal. Source resistance R_S is not considered for simplification of calculation, the circuit can be redrawn as shown below.

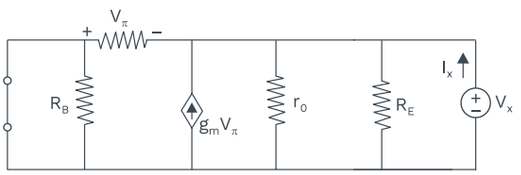


Fig. 4.42 Output Circuit of Common Collector Configuration

$$Z_0 = \frac{V_x}{I_x}$$

Applying KCL at output node, we get:

$$I_x + g_m V_\pi = \frac{V_x}{R_E} + \frac{V_x}{r_0} + \frac{V_x}{r_\pi}$$

Put $V_\pi = -V_x$

$$I_x - g_m V_x = V_x \left(\frac{1}{R_E} + \frac{1}{r_0} + \frac{1}{r_\pi} \right)$$

$$I_x = V_x \left(\frac{1}{R_E} + \frac{1}{r_0} + \frac{1}{r_\pi} + g_m \right) \text{ or } \quad \frac{V_x}{I_x} = \frac{1}{g_m} \| R_E \| r_0 \| r_\pi$$

$$Z_0 = \frac{r_\pi}{(1+\beta)} \| R_E \| r_0$$

- **Current gain (A_i)**

In the common collector configuration, the output current is I_e and input current is I_i.

Current gain, $A_i = \frac{I_e}{I_i}$

By current division rule, we get:

$$I_b = \left(\frac{R_B}{R_B + R_{ib}} \right) I_i$$

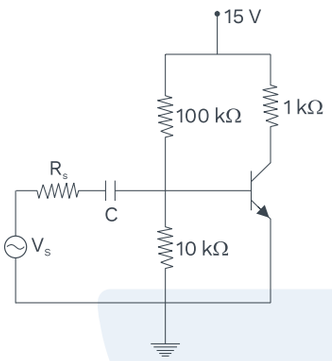
$$I_0 = (1 + \beta) I_b$$

$$I_0 = (1+\beta) \left(\frac{R_B}{R_B + R_{ib}} \right) I_i$$

$$A_i = \frac{I_0}{I_i} = (1+\beta) \frac{R_B}{R_B + R_{ib}}$$

Example 9:

A common emitter voltage divider circuit is shown below with transistor parameters β = 120 and V_{BE} = 0.7V, calculate (i) small-signal π-resistance and (ii) transconductance.



Sol:

For resistance and transconductance DC analysis is required and replace capacitor by open circuit.

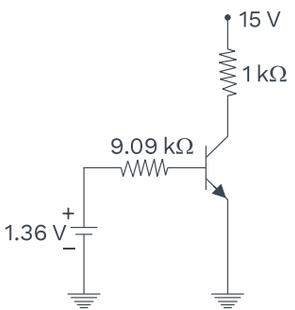
By Thevenin's equivalent,

$$V_{Th} = \frac{15 \times 10}{110}$$

$$V_{Th} = 1.36 \text{ V}$$

$$R_{Th} = R_B = \frac{100 \times 10}{110}$$

$$R_B = 9.09 \text{ k}\Omega$$



Applying KVL in the input loop, we get:

$$1.36 - 9.09I_B - 0.7 = 0$$

$$I_B = 0.0726 \text{ mA}$$

or $I_B = 72.6\mu A$

$I_C = \beta I_B$

$I_C = 120 \times 72.6\mu A$

$I_C = 8.71$ mA

i) Small-signal π resistance

$$r_\pi = \frac{V_T}{I_B}$$

$$r_\pi = \frac{26}{0.0726}$$

$r_\pi = 358.12\Omega$

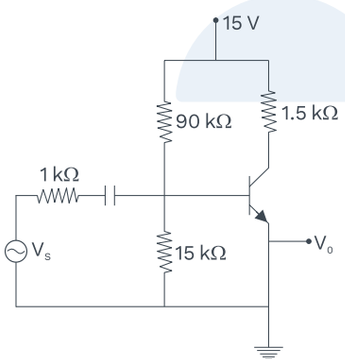
ii) Transconductance, $g_m = \frac{I_C}{V_T}$

$$g_m = \frac{8.71}{26}$$

$g_m = 335$ mA/V

Example 10:

A common emitter circuit is shown below. What is the voltage gain of the circuit?



Given, $\beta = 50$

(Assume output resistance of transistor r_0 to be very high)

Sol:

DC analysis:

- Replace capacitor by open circuit.
- Calculate the Thevenin's equivalent circuit of the voltage divider network.

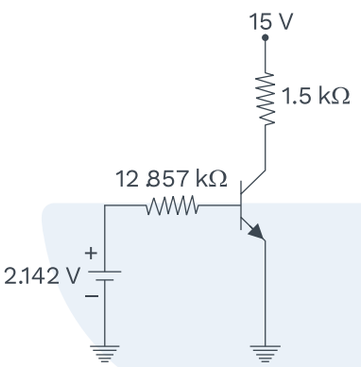
$$V_{Th} = \frac{15 \times 15}{105}$$

V_{Th} = 2.142 V

$$R_B = \frac{90 \times 15}{90 + 15}$$

R_B = 12.857 kΩ

$$r_\pi = \frac{V_T}{I_B}$$



Base current, $I_B = \frac{V_{Th} - V_{BE}}{R_B}$

$$I_B = \frac{2.142 - 0.7}{12.857}$$

I_B = 0.112 mA

$I_C = \beta I_B$

$I_C = 50 \times 0.112$

I_C = 5.6 mA

$$r_\pi = \frac{26}{0.112}$$

r_π = 0.23 kΩ

$$g_m = \frac{I_C}{V_T} = \frac{5.6}{26}$$

g_m = 215 mA/V

Voltage gain of the above circuit is given by

$$A_v = \frac{V_0}{V_s} = -g_m \frac{(R_B \,||\, r_\pi)}{(R_B \,||\, r_\pi + R_s)} (r_0 \,||\, R_c)$$

Since, r_0 is very high so:

$$r_0 || R_C \approx R_C$$

$$R_B \,||\, r_\pi = \frac{R_B r_\pi}{R_B + r_\pi}$$

$$= \frac{0.23 \times 12.857}{0.23 + 12.857} = 0.226 \text{ k}\Omega$$

Voltage gain,

$$A_v = -215 \times \left(\frac{0.226}{0.226 + 1} \right) \times 1.5$$

$$A_v = -59428 \approx -60{,}000$$

Chapter Summary



- BJT is mainly used for amplification of small AC signal.
- BJT cannot be used as an amplifier without proper DC biasing.
- The three most popular method for AC analysis of BJT is:
  - **i)** Hybrid equivalent model
  - **ii)** π-model (high-frequency model)
  - **iii)** r_e-model
- In DC analysis of BJT:
  - **i)** All AC voltage sources are replaced by short-circuit.
  - **ii)** All AC current sources are replaced by open circuit.
  - **iii)** Capacitors are replaced by open-circuit.
- In AC analysis of BJT:
  - **i)** All DC voltage sources are replaced by short-circuit, and all DC current sources are replaced by open-circuit.
  - **ii)** All capacitors are replaced by short-circuit.
  - **iii)** Small-signal equivalent circuit should be drawn with proper model.
- The important transistor parameters to be calculated from small-signal model are:
  - Voltage gain
  - Current gain
  - Input impedance
  - Output impedance
- Voltage gain is defined as the ratio of output voltage to the input voltage.
- Current gain is defined as the ratio of output current to the input current.
- The input impedance is the impedance seen from the input port. It is the ratio of V_i and I_i.
- The output impedance is the impedance seen from the output port. It is the ratio of V_0 and I_0.
- The input impedance of the BJT amplifier is not measured by an ohmmeter.
- The common emitter fixed-bias configuration provides significant voltage gain.
- In the common emitter configuration, output is taken across the collector.
- In the common base configuration, output is taken across the collector.
- In the common collector, output is taken across the emitter.

PRACTICE QUESTIONS

Q1 **Input impedance of BJT amplifier is the ratio of**
a) Input voltage and input current.
b) Output voltage and input current.
c) Input voltage and output current.
d) Output voltage and output current.

Q2 **Which of the following is the small-signal model used to represent an amplifier?**
a) Hybrid model
b) π-model
c) r_e-model
d) All of the above

Q3 **In a common emitter amplifier, the output is taken across:**
a) Base
b) Emitter
c) Collector
d) Any of the above

Q4 **The common collector amplifier is also known as:**
a) Base follower
b) Emitter follower
c) Collector follower
d) Cascode amplifier

Q5 **Which of the following configurations provides the highest power gain?**
a) Common base
b) Common emitter
c) Common collector
d) None of the above

Q6 When the DC analysis of the BJT amplifier is performed, the capacitor is replaced by

a) Short circuit
b) Open circuit
c) Capacitor is used as it is
d) None of the above

Q7 What is the effect of emitter resistance (R_E) on the input impedance?

a) It increases the input impedance.
b) It decreases the input impedance.
c) It increases the output impedance.
d) None of the above

Q8 Coupling capacitors are used to

a) Connect two circuits
b) Disconnect two circuits
c) Bias BJT
d) None of the above

Q9 What is the effect of the bypass capacitors on the voltage gain of an amplifier?

a) Increases
b) Decreases
c) No effect
d) First increases then decreases

Q10 Which of the following h-parameter is of amplifier equivalent to the current gain of an amplifier?

a) h_i b) h_f c) h_r d) h_0

Answer Key:

1)	a)	2)	d)	3)	c)	4)	b)	5)	b)
6)	b)	7)	a)	8)	a)	9)	a)	10)	b)